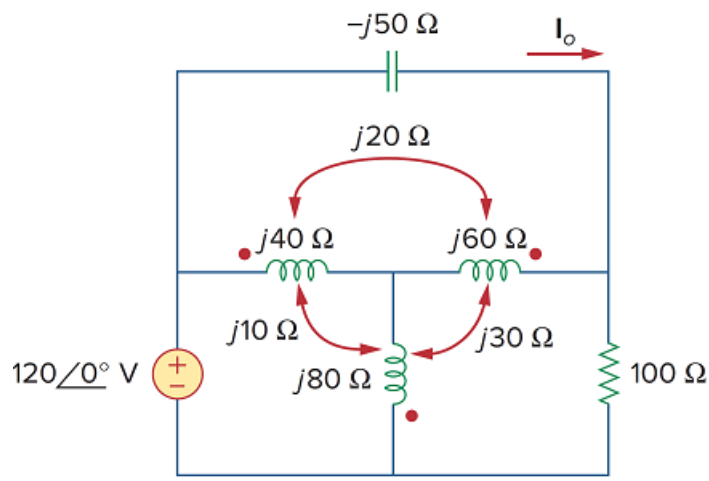


Problem

Find current I_o in the circuit of Fig. 13.91.

Figure 13.91



Step-by-step solution

Step 1 of 14

Refer to the Figure 13.91 from the textbook.

Assume the loop currents I_1 , I_2 & I_3 and the branch currents, i_1 , i_2 and i_3 as shown in the Figure 1.

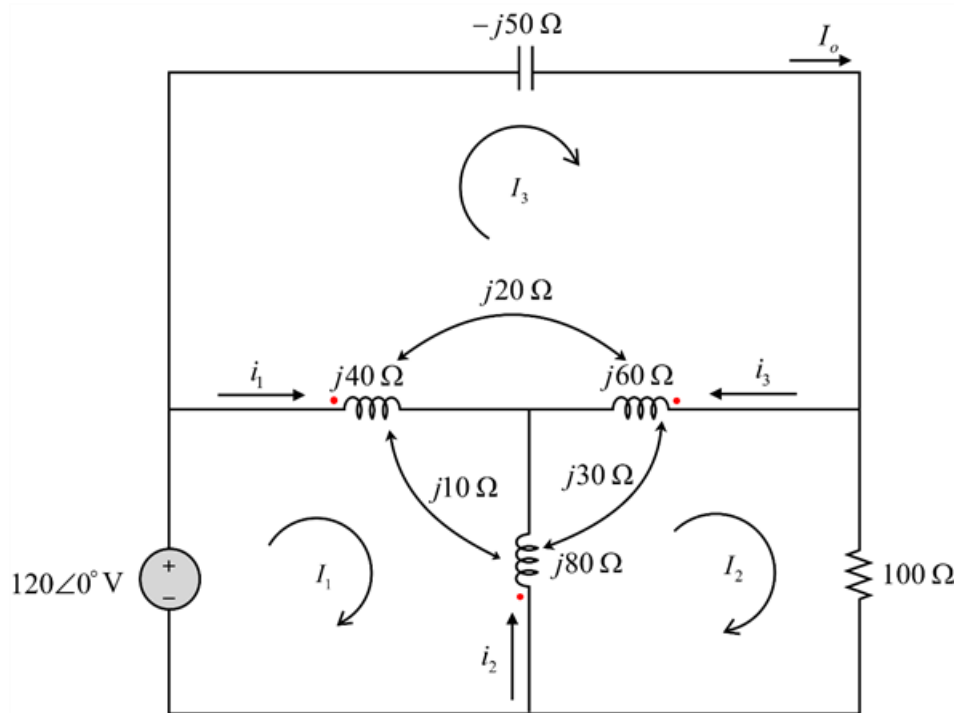


Figure 1

Step 2 of 14

Calculate the branch currents, i_1 , i_2 and i_3 .

$$i_1 = I_1 - I_3$$

$$i_2 = I_2 - I_1$$

$$i_3 = I_3 - I_2$$

Apply the dot convention and find the nature of connection of the coils.

Since in all coils, current enters at the dot, all coils are in series opposition with each other.

Step 3 of 14

Assume the mutual inductance between $j40\ \Omega$ and $j60\ \Omega$ to be M_{12} , between $j40\ \Omega$ and $j80\ \Omega$ to be M_{13} and between $j80\ \Omega$ and $j60\ \Omega$ to be M_{23} .

Thus,

$$M_{12} = M_{21} = j10\ \Omega$$

$$M_{23} = M_{32} = j30\ \Omega$$

$$M_{13} = M_{31} = j20\ \Omega$$

Step 4 of 14

Calculate the mutual voltage in $j40\ \Omega$ due to current, i_2 .

$$v_{12} = jM_{12}i_2$$

Substitute $j10\ \Omega$ for M_{12} .

$$v_{12} = j10i_2$$

Calculate the mutual voltage in $j40\ \Omega$ due to current, i_3 .

$$v_{13} = jM_{13}i_3$$

Substitute $j20\ \Omega$ for M_{13} .

$$v_{13} = j20i_3$$

Step 5 of 14

Calculate the mutual voltage in $j80\ \Omega$ due to current, i_1 .

$$v_{21} = jM_{21}i_1$$

Substitute $j10\ \Omega$ for M_{21} .

$$v_{12} = j10i_1$$

Calculate the mutual voltage in $j80\ \Omega$ due to current, i_3 .

$$v_{23} = jM_{23}i_3$$

Substitute $j30\ \Omega$ for M_{23} .

$$v_{23} = j30i_3$$

Step 6 of 14

Calculate the mutual voltage in $j80\ \Omega$ due to current, i_1 .

$$v_{31} = jM_{31}i_1$$

Substitute $j20\ \Omega$ for M_{31} .

$$v_{31} = j20i_1$$

Calculate the mutual voltage in $j80\ \Omega$ due to current, i_2 .

$$v_{32} = jM_{32}i_2$$

Substitute $j30\ \Omega$ for M_{32} .

$$v_{32} = j30i_2$$

Step 7 of 14

Sketch the modified circuit showing the mutual voltages as the voltage sources such that the coils are in series opposition as shown in Figure 2.

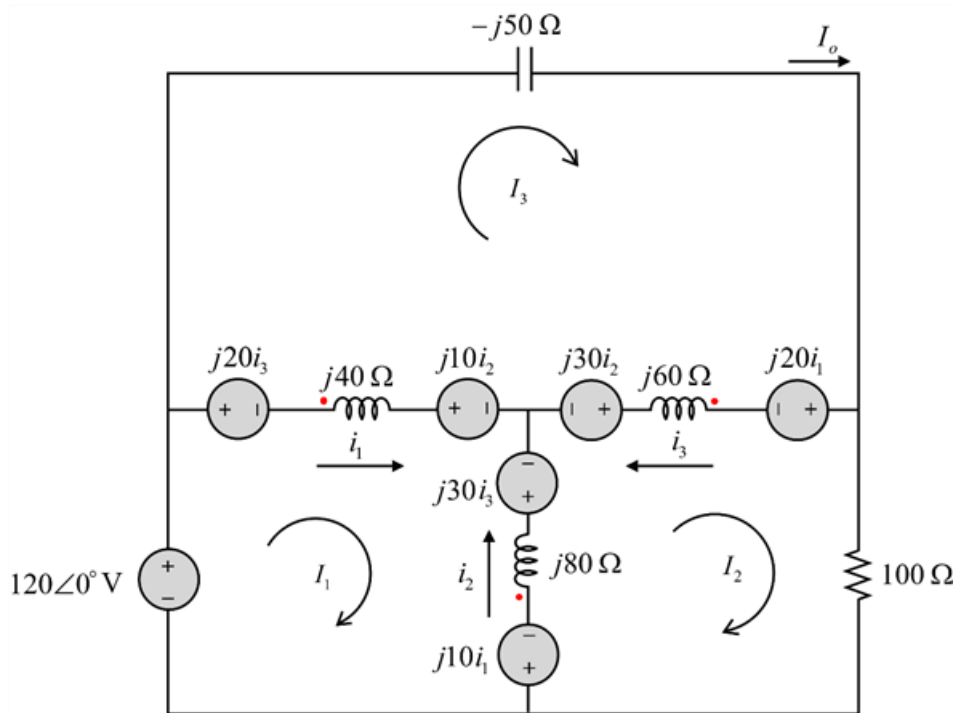


Figure 2

Step 8 of 14

Apply Kirchhoff's Voltage Law in the loop with the loop current, I_1 .

$$120 = j20i_3 + j40(I_1 - I_3) + j10i_2 - j30i_3 + j80(I_1 - I_2) - j10i_1$$

Substitute $I_1 - I_3$ for i_1 , $I_2 - I_1$ for i_2 and $I_3 - I_2$ for i_3 .

$$120 = j20(I_3 - I_2) + j40(I_1 - I_3) + j10(I_2 - I_1) - j30(I_3 - I_2) + j80(I_1 - I_2) - j10(I_1 - I_3)$$

$$120 = j(40 - 10 + 80 - 10)I_1 + j(-20 + 10 + 30 - 80)I_2 + j(20 + -40 - 30 + 10)I_3$$

$$120 = j100I_1 - j60I_2 - 40I_3$$

$$6 = j5I_1 - j3I_2 - j2I_3$$

Divide both the sides by j .

$$5I_1 - 3I_2 - 2I_3 = -j6 \dots\dots (1)$$

Step 9 of 14

Apply Kirchhoff's Voltage Law in the loop with the loop current, I_2 .

$$j10i_1 + j80(I_2 - I_1) + j30i_3 - j30i_2 + j60(I_2 - I_3) - j20i_1 = 0$$

Substitute $I_1 - I_3$ for i_1 , $I_2 - I_1$ for i_2 and $I_3 - I_2$ for i_3 .

$$0 = j10(I_1 - I_3) + j80(I_2 - I_1) + j30(I_3 - I_2) - j30(I_2 - I_1) + j60(I_2 - I_3) - j20(I_1 - I_3) + 100I_2$$

$$0 = j(10 - 80 + 30 - 20)I_1 + j((80 - 30 - 30 + 60) + 100)I_2 + j(-10 + 30 - 60 + 20)I_3$$

$$-j60I_1 + (100 + j80)I_2 - j20I_3 = 0$$

$$-3I_1 + (5 + j4)I_2 - jI_3 = 0$$

Divide both the sides by j .

$$-3I_1 + (4 - j5)I_2 - I_3 = 0 \dots\dots (2)$$

Step 10 of 14

Apply Kirchhoff's Voltage Law in the loop with the loop current, I_3 .

$$-j50I_1 + j20i_1 + j60(I_3 - I_2) + j30i_2 - j10i_2 + j40(I_3 - I_1) - j20i_3 = 0$$

Substitute $I_1 - I_3$ for i_1 , $I_2 - I_1$ for i_2 and $I_3 - I_2$ for i_3 .

$$0 = -j50I_3 + j20(I_1 - I_3) + j60(I_3 - I_2) + j30(I_2 - I_1) - j10(I_2 - I_1) + j40(I_3 - I_1) - j20(I_3 - I_2)$$

$$0 = j(20 - 30 + 10 - 40)I_1 + j(-60 + 30 - 10 + 20)I_2 + j(-50 - 20 + 60 + 40 - 20)I_3$$

$$0 = -j40I_1 - j20I_2 + j10I_3$$

$$-4I_1 - 2I_2 + I_3 = 0$$

Divide both the sides by j .

$$-4I_1 - 2I_2 + I_3 = 0 \dots\dots (3)$$

Step 11 of 14

Multiply equation (3) by $-\frac{3}{4}$.

$$3I_1 + 1.5I_2 - 0.75I_3 = 0 \dots\dots (4)$$

Add equations (2) and (3).

$$(4 - j5 + 1.5)I_2 + (-1 - 0.75)I_3 = 0$$

$$(5.5 - j5)I_2 - 1.75I_3 = 0$$

$$I_2 = \frac{1.75I_3}{5.5 - j5}$$

$$I_2 = (0.174 + j0.158)I_3$$

$$I_2 = 0.24I_3 \angle 42.27^\circ$$

Step 12 of 14

Multiply equation (3) by $\frac{5}{4}$.

$$-5I_1 - 2.5I_2 + 1.25I_3 = 0 \dots\dots (5)$$

Add equations (1) and (5).

$$(-3 - 2.5)I_2 + (-2 + 1.25)I_3 = -j6$$

$$-5.5I_2 - 0.75I_3 = -j6 \dots\dots (6)$$

Step 13 of 14

Substitute $(0.174 + j0.158)I_3$ for I_2 in equation (6).

$$-5.5(0.174 + j0.158)I_3 - 0.75I_3 = -j6$$

$$(-0.958 - j0.871 - 0.75)I_3 = -j6$$

$$(-1.708 - j0.871)I_3 = -j6$$

$$I_3 = \frac{-j6}{(-1.708 - j0.871)}$$

$$I_3 = 1.422 + j2.788$$

$$I_3 = 3.129 \angle 62.98^\circ \text{ A}$$

Step 14 of 14

Since the branch current, I_o is the same as the loop current, I_3 .

Thus,

$$I_o = I_3$$

$$= 3.129 \angle 62.98^\circ \text{ A}$$

Therefore, the current, I_o is $3.129 \angle 62.98^\circ \text{ A}$.